

Clustering of meteorological data to improve agricultural decisions: a case study with SIMAGRO-RS

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ABSTRACT

Context: Climate variability plays a fundamental role in shaping agricultural outcomes, influencing crop growth, yield, and resource management strategies. **Problem:** Identifying meaningful patterns in meteorological datasets to optimize agricultural practices is a challenging and complex task. **Solution:** This paper explores the use of clustering techniques on SIMAGRO-RS weather data to uncover meaningful patterns for agricultural applications. **IS Theory:** This study addresses the challenges of Information Systems applied to Agriculture, especially related to General Systems Theory (TGS). **Method:** The data was, first, extracted, then, preprocessed, filtered, aggregated, and standardized. Clustering was performed using K-means, K-medoids, and Hierarchical Agglomerative Clustering algorithms to cluster data from 19 meteorological stations in groups of stations with similar climate behavior. To define the number of clusters, the Elbow Plot and Silhouette Score methods were used. After clustering, the clusters were evaluated to identify the best result in terms of intracluster similarity. Supervised learning was used to design a specific metric for this problem, using an RMSE error measure. **Summary of Results:** From the experiments, the number of clusters $k = 2$ was found as the optimal number for the dataset. Regarding the clustering algorithm, the K-medoids algorithm presented better performance, with lower prediction RMSE (0.4990). **IS Contributions and Impact:** Based on the results, ways of interpreting clustering are suggested, in order to make it useful in the planning of agricultural practices in the state of Rio Grande do Sul.

CCS CONCEPTS

• **Computing methodologies** → **Cluster analysis**; • **Applied computing** → *Agriculture*; • **Information systems** → *Decision support systems*; *Data mining*; Sensor networks; • **Mathematics of computing** → *Time series analysis*.

KEYWORDS

Clustering, Time Series, Meteorological Data, Agriculture

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1 INTRODUCTION

Agriculture is intrinsically related to climatic conditions, and understanding the relationship between weather patterns and agricultural outcomes is really important. Weather monitoring and alert systems deployed in agricultural regions offer comprehensive repositories of meteorological data, providing a good opportunity to investigate the interconnections between weather and agriculture [2]. Traditional analyzes often struggle to capture the complexity of these relationships due to the multidimensionality and temporal variability of weather data [20]. Clustering, a data-driven approach, has emerged as a powerful tool to unveil underlying structures within datasets, aiding in the identification of distinct agroclimatic regions, and facilitating the use of site-specific management practices. Thus, computer systems are essential and play a more prominent role in the development of solutions for this context [18].

In Brazil, there are efforts to develop solutions to combat climate change, such as MapBiomass project [30], widely recognized for having an extensive database and a very mature platform for visualizing maps with land use data, climate variables and many functionalities. In this direction, but with a more specialized scope

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and focused on regional agricultural practices, some efforts seek to assist producers, not only with climate data, but also with useful insights for different seasons and crops [3, 25].

In the state of Rio Grande do Sul (RS), the southernmost state in Brazil, which has a largely agricultural economy, SIMAGRO-RS [13] is one project that has emerged in the context of climate monitoring. With weather stations deployed in different locations in RS, SIMAGRO-RS collects data and monitors the state's climate, providing information through an online platform to help the agricultural sector with planning activities and guide short, medium, and long-term actions. As this system is quite recent and with the deployment of meteorological stations still in progress, the data produced by SIMAGRO-RS are starting to be used in climate studies in the state of RS, providing a great opportunity for research such as the one presented here.

This study investigates the application of clustering techniques to multivariate time series that represent meteorological observations collected by SIMAGRO-RS. By leveraging clustering methodologies, the purpose is to uncover latent agroclimatic zones and enhance decision making for agricultural practices. The main objective of this study is to deal with the SIMAGRO-RS weather dataset to cluster stations with similar characteristics regarding meteorological variables of interest to agriculture. This paper represents an experimental stage of a broad doctoral thesis research and extends initial results presented in [10], going beyond by increasing the number of investigated climate variables (Temperature, Relative Humidity, Rainfall, and Solar Radiation), exploring different algorithms for clustering the multivariate time series (K-means, K-medoids, and Hierarchical Agglomerative Clustering), developing a context-specific metric, and directing the cluster analysis to the agricultural perspective in the state of Rio Grande do Sul. To achieve this, the study included the following specific contributions:

- (1) Preprocess data collected at SIMAGRO-RS weather stations to filter, quality check, aggregate, and standardize for clustering.
- (2) Cluster SIMAGRO-RS meteorological stations from an agrometeorological perspective, investigating different algorithms and evaluating the results based on the climatic characteristics of each region;
- (3) Implementing a software package based on unsupervised ML to preprocess and cluster SIMAGRO-RS weather data.

The clustering presented in this work grouped 19 SIMAGRO-RS meteorological stations according to the processes described below. This grouping respects the historical division of the political and economic regions of the RS, while presenting its main contribution in adjusting the limits between the regions, according to the data collected in the last four years. Based on the results and the software package implemented and made available, we seek to enable more researchers to complement and advance research on this topic.

In the scope of Information Systems, this paper deals with challenges of Information Systems applied to Agriculture [32] and it can be effectively related to the General Systems Theory (GST) [34] due to its emphasis on a holistic perspective and the interrelation of components within a larger system. The General Systems Theory posits that systems share common principles and structures, and their behavior is governed by the interactions between

their elements. In the agricultural context [17], as is the case with this study, the proposed clustering methodology, along with the decision-making improvements, can be viewed as interconnected components of a broader agricultural system [15].

The article is structured as follows: Section 2 presents Related Works found in the literature, Section 3 presents the SIMAGRO-RS project and the characteristics of the dataset used in this work, Section 4 presents the Methodology used in the work and details the steps taken, Section 5 presents the Results and Discussion, and Section 6 presents the Conclusion and Future Works.

2 RELATED WORKS

Meteorological data can be applied for different purposes in the agricultural context. When used in clustering tasks, they are generally related to the identification of patterns, such as the regionalization of some agricultural characteristic. Below are some examples and a brief discussion of results and challenges of related applications involving clustering of climate data.

In [4], hydrometeorological data from the state of Rio Grande do Sul were analyzed to identify watersheds with similar hydrological behavior. Aggregated variables were used in annual averages, such as precipitation and water flow, representing more than 30 years of measurement, in which hierarchical and non-hierarchical clustering techniques were used to delimit hydrologically homogeneous regions in the state of Rio Grande Sul. The clustering proposed by the authors, as the main outcome of the work, aims to help researchers choose the best method to be used in regionalization studies in Rio Grande do Sul, and contributes to better calibration of hydrological forecasting models and decision making.

The study presented in [1] aims to highlight climate patterns that influence the interannual variability of rice productivity in Rio Grande do Sul (RS). Hierarchical and non-hierarchical clustering methods were used, and the analyzed dataset contained rice productivity data from RS in the period between 1990 and 2013. Correlations were observed between climate indicators and productivity, as well as correlations with climatic phenomena such as ENSO (El-Niño Southern Oscillation).

The work presented in [29] aims to group regions with similar climates in Brazil, identifying regional climatic specificities. The authors use the method of Grade of Membership (GoM) and the meteorological variables Rainfall, Relative Humidity, Maximum and Minimum Temperature, and Wind Speed, obtained between January 1981 and December 2013. The work brings unprecedented and very relevant results for Brazil, in general, but does not specialize the study for the states.

With a similar approach, but with data from different sources, in the work presented in [7] the authors used climate and NDVI data from the MODIS and GIMMS databases, in image and time series format, to classify homogeneous regions of vegetation cover in Rio Grande do Sul. The algorithm used to cluster the data was K-means. The temporal dynamics presented a typical seasonal pattern, with amplitude variations, which allows monitoring phenological changes and enables studies of land cover change, making it possible to identify and map the main vegetation cover that occurs in RS.

Related works show the viability of research in the direction of grouping climate data, although there are few that deal with

current and specific data from Rio Grande do Sul. Also, it is noted that in the area of Computer Science there is still a gap to be filled with the development of research relating climate data and machine learning techniques, such as clustering. This shows that there is a good space for research on this subject, especially with regard to the specificities of different agricultural areas, as is the case with the SIMAGRO-RS database. Thus, in order to contemplate these needs and contribute to the development of the base activity of the economy in Rio Grande do Sul, this work aims at building knowledge from the data generated by SIMAGRO-RS and pushing the frontier of the state of the art in this context.

3 SIMAGRO-RS DATASET

SIMAGRO-RS (Sistema de Monitoramento e Alertas Agroclimáticos)¹ is the Rio Grande do Sul Agroclimatic Monitoring and Alerts System and is a comprehensive and dynamic online platform, with meteorological maps and a repository of RS weather data. Started in 2019 and founded on the principle of providing historical and real-time climate information, SIMAGRO-RS serves as an invaluable tool for understanding the intricate interplay between weather dynamics and agricultural activities. The database captures a diverse array of more than twenty meteorological variables, every half hour, such as temperature, dew point, precipitation, wind speed, wind direction, solar radiation, and relative humidity, collected through an extensive network of meteorological stations distributed across the state. Figure 1 shows an example of data presentation on the SIMAGRO-RS platform, through the state map with pins representing the weather stations. The platform also has options for viewing data in table format, as well as extracting data in formats such as .csv, .pdf and .xls. With its expansive coverage and high-frequency data collection, SIMAGRO-RS offers a unique opportunity for researchers, producers, and policy makers to gain insight into the agroclimatic conditions that significantly influence crop growth, yield, and overall agricultural productivity.

The SIMAGRO-RS project is an initiative of the RS state government, through the Department of Agriculture, Livestock, and Rural Development (Secretaria de Agricultura, Pecuária e Desenvolvimento Rural do RS), and beyond its role as a repository of meteorological data, provides a critical foundation for the advancement of precision agriculture and sustainable resource management. The database enables the characterization of local climate variations, the identification of microclimates, and the monitoring of climate trends over time. By offering a granular view of weather patterns on regional and local scales, SIMAGRO-RS empowers agricultural stakeholders with the information needed to make informed decisions related to crop selection, planting times, irrigation strategies, pest control measures, and other vital aspects of agriculture. As the agricultural sector faces increasing challenges posed by climate change and fluctuating weather patterns, the insights drawn from SIMAGRO-RS are poised to play an instrumental role in shaping resilient and adaptive agricultural practices that can secure food production in the face of a dynamic climate landscape.

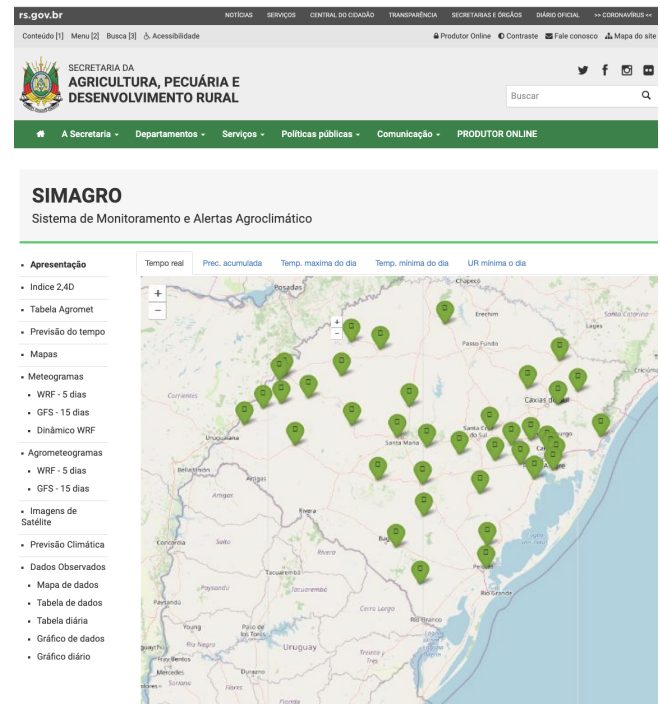


Figure 1: SIMAGRO-RS web interface, with a map of meteorological stations throughout the state.

4 METHODOLOGY

This section describes the Methodology followed in this work. Initially, the data extraction process is detailed. Then, the data preprocessing steps are presented, and also the clustering implementation as well. All the files and code implemented are available in the project repository on GitHub.²

An overview of the processes in this study is presented in Figure 2. First, the raw data were extracted from SIMAGRO-RS in text files and imported into Python notebooks, for initial treatment. Then, the data were preprocessed, starting with filtering, according to the variables and the time period to be analyzed. The preprocessed data were then aggregated into representative daily values, resulting in a multivariate time series for each weather station. After aggregation, the data were standardized, in order to transform the data into a representation of all variables on the same scale, for better clustering performance. Then, the resulting time series were used as input to the search for an optimal number of clusters and clustered using the K-means, K-medoids, and Hierarchical Agglomerative Clustering algorithms. Finally, the clustering result was evaluated to identify the best performance and suitability for this case study. All of these intermediate processes are detailed in this section.

4.1 Dataset Extraction

For this study, weather data were first extracted from the SIMAGRO-RS database in .txt files. Data were extracted from 19 meteorological stations, which represent the first stations installed since

¹SIMAGRO-RS — <http://www.simagro.rs.gov.br/>

²GitHub Rep. - <https://github.com/marcosjr06/ClusterSimagroRS>

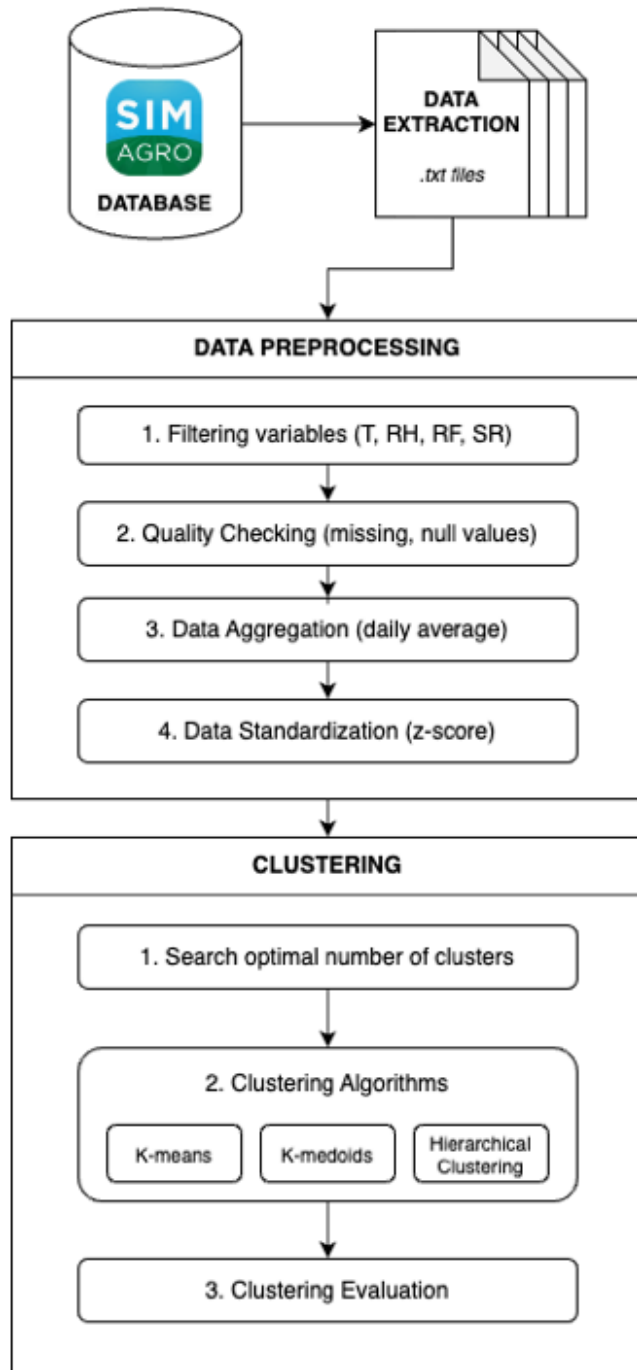


Figure 2: Methodology overview: sequence of steps for data extraction, preprocessing and clustering.

Table 1: SIMAGRO-RS Meteorological Stations used in this study.

ID	Station (City)	Latitude (Deg)	Longitude (Deg)
1	Pinheiro Machado	-31.577629	-53.383749
2	Piratini	-31.447456	-53.103847
3	São Sepé	-30.171961	-53.571799
4	Itaqui	-29.129403	-56.551439
5	Maçambará	-29.147451	-56.067150
6	Rosário do Sul	-30.250008	-54.916827
7	Getúlio Vargas	-27.888607	-52.228262
8	Ilópolis	-28.926757	-52.126125
9	Barra do Ribeiro	-30.299519	-51.304567
10	Cachoeira do Sul	-30.032233	-52.893182
11	Canguçu	-31.398944	-52.677631
12	Herval	-32.028922	-53.394227
13	Lavras do Sul	-30.810365	-53.899211
14	São Borja	-28.659901	-56.009982
15	Bossoroca	-28.731386	-54.903388
16	Jaguari	-29.491148	-54.689576
17	Caxias do Sul	-29.161301	-51.191962
18	Porto Vera Cruz	-27.734706	-54.899875
19	Sobradinho	-29.415668	-53.027771

the beginning of the project. Currently, more stations are in operation, however, as the project is recent and installations occur asynchronously, many of the stations are quite new, implemented in the current year, and, consequently, do not have the historical data needed for clustering tasks. Table 1 presents basic information on the 19 meteorological stations used in this study.

4.2 Data Preprocessing

In order to prepare the data to be used in clustering tasks, it was necessary to perform some pre-processing steps, such as filtering, quality checking, aggregation, and standardization. These steps are described below.

4.2.1 Filtering. After extraction, the data were filtered by desired variables and the period of time to be observed. To select the variables of interest, the relevance in agricultural production of RS crops was considered, and the variables Temperature (T), Relative Humidity (RH), Rainfall (RF), and Solar Radiation (SR) were selected. Regarding the time period, the agricultural production season in the state of RS was considered, for the main grain crops, such as soybeans, which mainly comprise spring and summer. Therefore, data from 182 days, between October 1 and March 31, of the last three seasons of agricultural production (2020-2023) were selected.

4.2.2 Quality Checking. With regard to quality checking, a scan for missing and null values was performed. For the 19 stations used in this study, 25044 missing values and 8428 null values were detected, including all variables, which represent about 6.7% of the dataset. Given the high frequency of sensing and the ability to use aggregated values, it was decided to disregard these missing and null values, and aggregate all variables into daily values.

4.2.3 Aggregation. Data were aggregated into daily values, considering good aggregation practices for each variable in agricultural context. Initially, for the variables T and RH, daily averages were used, while for RF and SR, daily accumulated values were used. After this first aggregation, given the three seasons under analysis (20/21, 21/22 and 22/23), the average values of T, RH, RF and SR were calculated for each day of these seasons, in order to identify a single value for each day of the season that represents the data collected in the last three years. There is a data gap for stations 14 to 19, in the first two months (October and November) of the first season (20/21). For these stations, in this period, only data from the second and third seasons were used to compute the aggregated values. Figure 3 presents the aggregated data for the four variables of this study, separately (first graph represents the Temperature, in black; second graph represents the Relative Humidity, in green; third graph represents Rainfall, in blue; and fourth graph represents Solar Radiation, in orange), for meteorological station number 10, in the city of Cachoeira do Sul. In this image it is possible to observe the values aggregated daily (during the 182 days, between October and March) for the last three seasons of agricultural production.

4.2.4 Standardization. Standardization is an essential step to ensure that all features are on the same scale, which can improve the performance of clustering algorithms and make them more robust. Can be done in different ways when applied to machine learning, basically transforming the variables in the dataset. In this study, the *z-score* [14] technique was used to transform the variables so that they have a mean of 0 and a standard deviation of 1. The *StandardScaler* function was used from the *scikit-learn* [23] Python library. Figure 4 shows the distribution of variables, plotted on the same scale, after standardization. The vertical axis on each of the graphs in Figure 3 (with different measurement units) and on the graph in Figure 4 demonstrates the scale transformation of the variables.

4.2.5 Clustering dataset. After all the preprocessing steps, the dataset prepared for clustering was composed of 19 standardized multivariate time series (one for each meteorological station) of the same size, containing 182 timestamps each (days). In each timestamp, the time series stores 4 values, representing the climatic variables used in this study. All preprocessing steps were implemented in Python, using auxiliary data manipulation libraries such as the well-known *pandas*, and compose the software package available in the GitHub repository.

4.3 Clustering Implementation

The clustering process was implemented in Python and the code produced in this step is also published with the software package on the project's GitHub repository. For clustering tasks, the auxiliary libraries *scikit-learn* [23], *ScyPy* [33] and *tslearn* [31] were used mainly, as they are very useful and powerful libraries to deal with time series and uncover temporal patterns and relationships within meteorological datasets. These libraries implement algorithms for clustering time series, and some complementary adaptations were implemented to deal with multivariate series. Basically, first, it is necessary to define the number of clusters to be searched, and, later, the execution parameters. Then, an evaluation

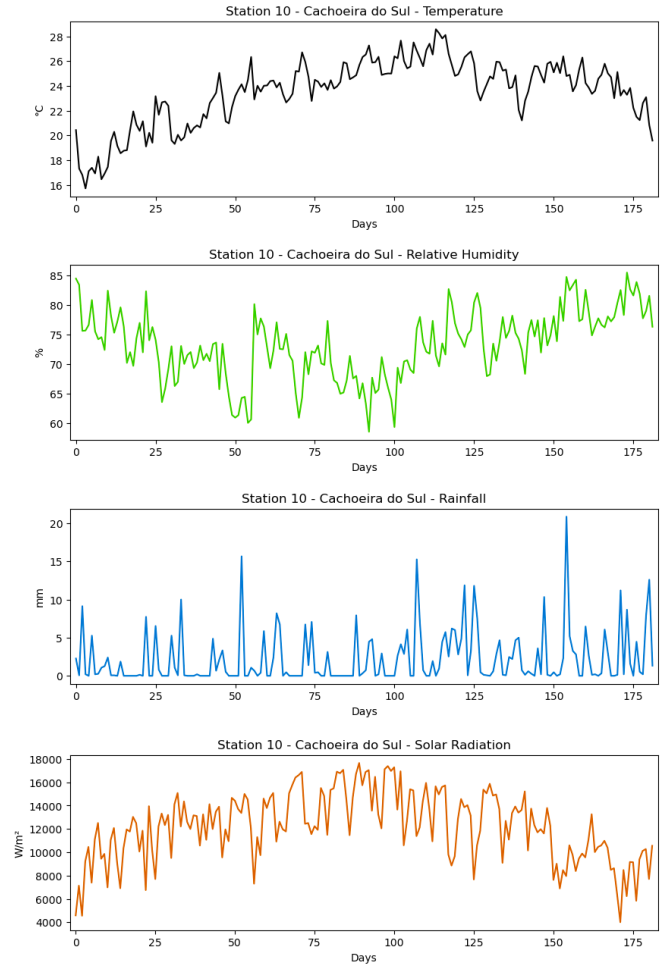


Figure 3: Visualization of the variables of interest in this study (Temperature - black; Relative Humidity - green; Rainfall - blue; Solar Radiation - orange), separately.

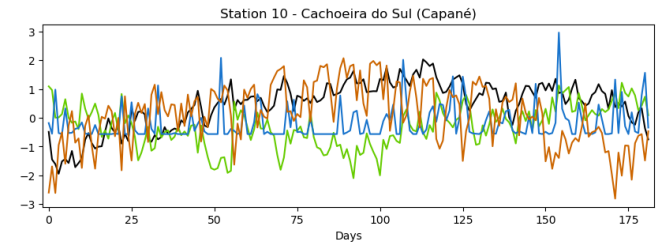


Figure 4: Data distribution after standardization (Temperature - black; Relative Humidity - green; Rainfall - blue; Solar Radiation - orange).

of the clustering results was made. The details of these steps and the implementation are described below.

4.3.1 Number of clusters. The choice of the number of clusters, often denoted k , is a critical decision when instantiating clustering algorithms. The number of clusters directly influences the granularity and interpretability of the clustering results. In general, the selection of the number of clusters requires a combination of computational methods and contextual understanding to ensure meaningful results. To determine the optimal number of clusters, several techniques can be employed as follows.

The *Elbow Method* [8], one of the most common methods for this task, involves running the clustering algorithm for a range of values of k and plotting the sum of squares (WCSS) or a similar metric against the number of clusters. The plot often exhibits a characteristic *Elbow Point* where the reduction in WCSS slows down, indicating that the addition of more clusters does not significantly improve the fit of the model. The elbow plots generated for the algorithms investigated in this work are available in the GitHub repository, due to space limitations here. Analyzing the graphs resulting from the Elbow Plot computation, they were not sufficiently conclusive regarding the optimal number of clusters. This is common when searching for an optimal number of clusters, due to the difficulty of a single metric accurately defining this. Therefore, another metric was also used in a complementary way.

The other method used was the *Silhouette Score* [26], which quantifies how similar an object is to its own cluster compared to other clusters. The silhouette score can vary between -1 and 1, a higher score indicates better-defined clusters. The search for the Silhouette Score in this study was performed in the range of 2 to 10, in order to try some possibilities of the number of clusters, within a viable limit for the agricultural context and the number of stations in use. Table 2 presents the scores for each algorithm and cluster number, with the highest score highlighted for each algorithm. Based on these results, analyzing the *Elbow Plot* and the *Silhouette Score* in a complementary way, the optimal number of clusters for this study with the SIMAGRO-RS dataset was defined as $k = 2$.

4.3.2 Cluster Algorithms. With the optimal number of clusters defined, other parameters were configured, such as the seed number, the number of iterations, and a metric to calculate the distances between time series. The implementation details, as well as the metrics for calculating the distances between the time series, are detailed in the code on GitHub, specifically for each algorithm.

Table 2: Silhouette Score for SIMAGRO-RS dataset.

N of clusters	K-means	K-medoids	Hierarc. Clust.
2	0.420	0.231	0.217
3	0.218	0.130	0.130
4	0.263	0.147	0.133
5	0.202	0.073	0.127
6	0.235	0.044	0.131
7	0.198	0.043	0.136
8	0.187	0.065	0.126
9	0.124	0.035	0.124
10	0.190	0.022	0.125

As input for clustering, the 19 standardized multivariate time series, produced as output in the data preprocessing stage, were used. The clustering was then run 10 times and the average cluster across all runs was calculated. The algorithms used were K-means, K-medoids, and Hierarchical Agglomerative Clustering and are introduced below. The clustering results are presented in the next section.

- (1) *K-means*: a clustering algorithm that partitions a dataset into k clusters by iteratively assigning data points to the nearest centroid and updating centroids based on the mean of the assigned points [21]. The choice of k , the number of clusters, and the sensitivity to initialization and scalability are challenging. Multiple initialization runs and alternative distance metrics can help mitigate these issues. In this study, a random number was used as seed number, with 10 initializations, and the chosen metric for cluster assignment was *Dynamic Time Warping (DTW)* [28] barycenter average.
- (2) *K-medoids*: akin, this algorithm partitions a dataset into k clusters. However, *K-medoids* use the data points themselves, known as medoids, as representatives of clusters [27]. This approach makes *k-medoids* particularly useful in scenarios where the mean may not accurately represent the central tendency of a cluster. In this study, the standard parameters of the algorithm were used, with *Euclidean* distance as cluster distance metric.
- (3) *Hierarchical Agglomerative Clustering*: a clustering technique that builds a hierarchical representation of data by iteratively merging clusters [22]. This algorithm can produce dendrogram structures that display hierarchical relationships among clusters. Agglomerative clustering starts with each data point as its own cluster and iteratively merges the closest clusters until a single cluster, containing all data points. The choice of linkage criterion and distance metric influences the clustering results. In this study, the *Ward* variance minimization metric was used.

4.3.3 Cluster Evaluation. Evaluating the quality of clustering is fundamental to assess the meaningfulness of generated clusters and to select algorithms, adjust parameters, and interpret results. Various metrics and techniques have been developed for this purpose, encompassing internal, external, and relative criteria. The choice of evaluation method should be guided by the specific goals of the analysis and the characteristics of the data, as no single metric provides a comprehensive view. Internal metrics, such as the silhouette score, assess the quality of clusters based on the intrinsic properties of the data and the clustering results themselves. In this case, where we already have silhouette scores (Table 2), it is observed that although we have chosen the optimal number of clusters, the quality of the clusters is average, as the scores are closer to 0 (average) instead of 1 (best score).

In addition, specific evaluation techniques can be implemented to increase the reliability and robustness of the results. For this study with SIMAGRO-RS data, a specific metric was developed to evaluate the quality of the clusters, in accordance with the general purpose of intracluster similarity sought in this context. Since we are looking for clusters that represent the same climate zone, an evaluation metric can be based on supervised learning, with the

purpose of predicting data according to data from a cluster and evaluating how good this prediction is.

This metric works like this: from a cluster C , composed of s stations ($s_1, \dots, s_n$), a station s_i is removed and a supervised learning method, such as Linear Regression, is used to predict the values of s_i using the remaining stations in the cluster C . Then, the *RMSE (Root Mean Squared Error)* of the prediction is calculated, comparing the real data with the predicted data. The prediction is then computed for all stations, through this process of removing it from the cluster, predicting it, and re-adding it to the cluster, and each station had an assigned RMSE. Then, an average RMSE can be computed for cluster C , based on the RMSEs of stations $s_1, \dots, s_n$. With the RMSEs of each cluster, the average RMSE of that entire clustering is calculated. The algorithm that had the lowest Total RMSE was considered the best, as this indicates a greater similarity between the stations in those clusters. The results of the clustering assessment of this study are presented in the following section.

5 RESULTS AND DISCUSSION

This section compiles the main results found after the clustering of SIMAGRO-RS data, as well as their evaluation, interpretation, and a discussion on possibilities of how to use the proposed clustering.

5.1 Clustering Results

After the clustering was performed, the results were consolidated and are presented below. As detailed in the previous section, the optimal number of $k = 2$ clusters was found, and the K-means, K-medoids and Hierarchical Agglomerative Clustering algorithms were used to group the SIMAGRO-RS meteorological stations. Due to space limitations, the complete visual result is presented in the GitHub repository and a general summary is shown here.

The basic result of clustering is the spatial representation, on the map, of stations divided into their clusters. Regarding the clustering itself, the results were quite similar, grouping the 19 stations into 2 clusters, differing in the definition of the border between the clusters. Figure 5 shows the clustering obtained by the K-medoids algorithm, which was considered the best performance for this study. The K-means cluster differs from this only by placing two stations (*Sobradinho* and *Lavras do Sul*) in the eastern cluster, in green, while Hierarchical Clustering placed only one station (*Lavras do Sul*) in the green cluster instead of the red one.

The definition of best performance was based on the metric developed for this specific study, given the context, as detailed in the previous section. The average RMSE of the clusters was calculated, using linear regression to predict data based on its own cluster. Table 3 presents the result of this metric for each cluster and algorithm, as well as the Total RMSE, with the best value highlighted. It is possible to observe that the clustering with the K-medoids algorithm obtained a lower error, therefore, being considered a more appropriate grouping of these weather stations.

It is possible to observe that the clustering follows a spatial pattern, grouping adjacent stations, and establishing the boundaries between one cluster and another. This is a behavior that makes sense when it comes to weather patterns. One cluster encompasses stations in the center and west of the state (red circles), and another encompasses the eastern and southern regions (green squares).

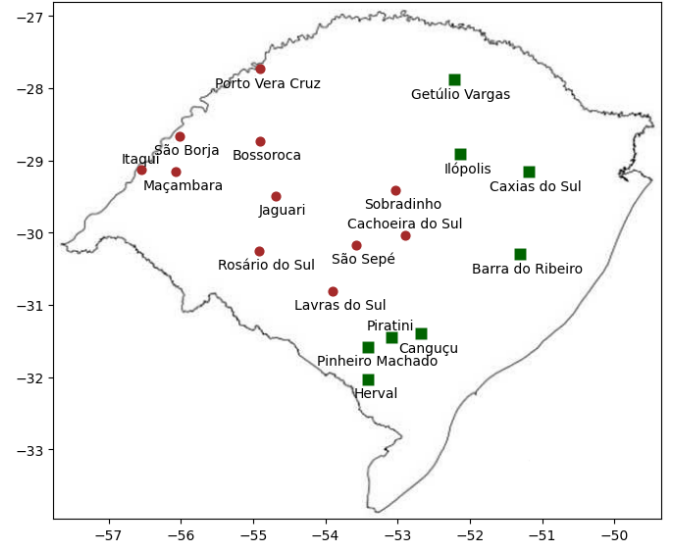


Figure 5: Clustering results of the SIMAGRO-RS dataset with algorithm K-medoids and number of clusters $k = 2$, one cluster in green squares, another in red circles.

Table 3: RMSE of cluster-based prediction.

Cluster	K-means	K-medoids	Hierarc. Clust.
C1 RMSE	0.5091	0.4887	0.5067
C2 RMSE	0.5310	0.5092	0.5164
Total RMSE	0.5201	0.4990	0.5115

This division makes sense compared to the traditional division of microregions in the state of Rio Grande do Sul [11], merging traditional climatic regions of the state, possibly due to the low number of variables and the few years observed. What appears to be the main contribution of this clustering, compared to the traditional division of microregions in RS, is a grouping of microregions into larger groups and the movement of the limits of these regions, adjusting the climatic zoning according to the meteorological data of the last three years.

Furthermore, a relevant contribution of Hierarchical Clustering is that the resulting dendrogram shows the relationships between weather stations, as shown in Figure 6, highlighting the strongest similarities in climate patterns. Despite the station being placed in a different cluster (*Lavras do Sul*), which would deserve a more in-depth analysis, for the other stations it is possible to observe the strongest relationships between locations, which can provide very interesting information for agriculture. For example, in an agricultural activity where there is no clear definition of the climate situation for a location, this dendrogram can serve as a guide to where to look for similar conditions, helping to form the basis for making a decision.

In general, the main limitation of this work is the still low availability of data. Of course, given the limited amount of data from just three seasons, the analysis is insufficient to determine whether

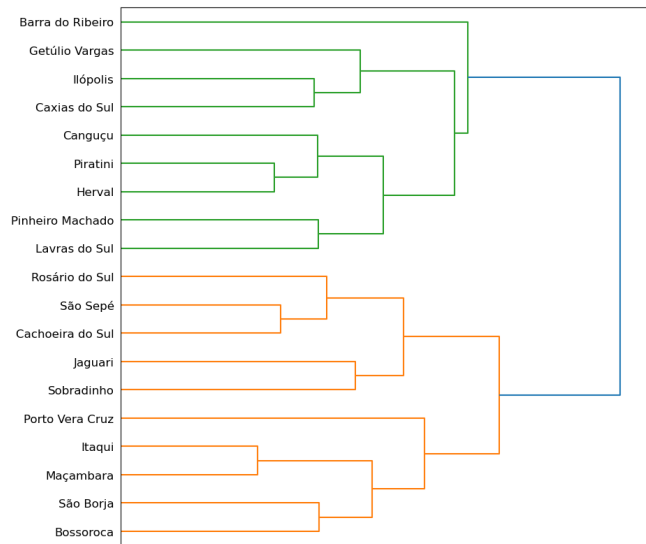


Figure 6: Dendrogram resulting from Hierarchical Agglomerative Clustering.

this clustering pattern is the result of recent years only or has been occurring for longer. However, knowing the current situation of climate change occurring in different places around the world, monitoring data like these can help to identify changes more quickly, adjusting production patterns and contributing to accuracy in agricultural practices. Below are listed some possible uses of this proposed clustering.

Furthermore, in relation to the application of ML techniques, a possible advance would be in the automation of the choice of the clustering algorithm, based on an AutoML approach, for example, which has been applied to the agricultural context [12]. The process would involve applying all methods multiple times and storing the best results. In the long term, the results obtained could feed a decision mechanism, also developed with ML, for choice. Although, since the data volume for this needs to be relatively large and the proposed clustering process is not executed in a specially optimized computing environment (e.g. HPC devices), at first, this possibility was not feasible within the scope of developing this paper. The case study presented here reflects an initial state of investigation, and also for practical reasons, we understand that it is not convenient for now to choose an algorithm automatically, since the performance metrics used and analysis of the results still partially rely on empirical knowledge of the agricultural stakeholders involved. This is an important feature and challenge to be considered for the evolution of work in a more robust decision system.

5.2 Discussion

From the Information Systems perspective, the results denote the importance of the system and the application of theory to assess how agricultural innovations increase the capacity of agricultural systems to adapt to climate change. As it is a case study, the focus at this time was to execute an initial set of algorithms on an initial set of data and evaluate performance, but this artifact aims

to serve as a basis for the evolution of research in the format of a framework, based on the processes presented in Figure 2. Therefore, this research aligns with the GST by recognizing the interconnectiveness of various components involved in enhancing agricultural decisions.

The GST allows for the examination of how patterns in meteorological data can affect the agricultural decision-making process, considering both internal elements (algorithms) and external elements (agricultural practices). In this study GST was used to understand how innovation in agricultural technologies and practices can increase the adaptability of agricultural systems to climate variations. The proposed methodology can be evolved into an agricultural decision support system, with a well-defined flow of operations. The process presented, and implemented in the form of a software artifact, reflects a system that includes a database, data manipulation, clustering and, at the end, a graphical visualization of the result obtained. This software artifact is composed of different components that can be extended to include access to different data sources (databases with meteorological data), different classification methods (ML techniques) and way of visualizing results (different ways of presenting data), in the same direction of related works with different purposes [16, 24].

Regarding the relationship, integration and exchange of information with other platforms, at this first moment there is still no direct action with other platforms, which is understood to be perfectly viable. In this case study, the focus was on the treatment of data obtained by SIMAGRO, in order to extract new layers of knowledge from the sensed variables, creating knowledge from the monitored data, but at an early stage. Possible forms of relationship between the platforms are being investigated, such as combining data from platforms to improve clustering, especially those that have been in operation for longer, while SIMAGRO-RS has only been active for a few years.

Certainly, data and tool integration processes would involve many complex challenges, such as data cross-referencing issues. Addressing heterogeneity when cross-referencing data is essential for establishing a cohesive and reliable information framework. In order to effectively manage diverse data sources, it is crucial to implement, for example, standardized formats and thorough data cleaning processes. This corroborates the importance of standardization, adopted in the methodology. Standardization ensures a common ground for comparison, while cleaning procedures tackle inconsistencies and variations in the data. Also, employing data integration techniques, such as creating a data warehouse or utilizing ETL processes, helps unify disparate datasets into a cohesive structure. Additionally, a flexible, scalable architecture are integral components for successfully navigating heterogeneity in data cross-referencing, ultimately supporting robust decision-making systems, especially in fields like agriculture.

5.3 Main Outcomes

The results obtained here can be used in several analyses, within the agricultural context. As main outcomes of this study, the following possibilities and research directions stand out.

- **Agroclimatic Zoning [5]:** use the proposed cluster to specialize agricultural strategies and to segment the state into micro

agroclimatic zones, each characterized by coherent weather patterns. This zoning can provide a better understanding of microregional climate variations, enabling the tailoring of crop choices and cultivation practices to specific zones.

- Crop-Specific Insights [19]: use this cluster to uncover weather conditions that are most conducive to the growth and development of specific crops. By linking these clusters to the stages of crop growth, farmers can make informed decisions regarding planting, irrigation, and pest management.
- Extreme Weather Event Detection [6]: based on this cluster, it can be possible to identify anomalous weather events, enabling early warning systems to mitigate the impact of extreme conditions on crops. This can help improve agricultural resilience and minimize yield losses.
- Decision Support for Agriculture [35]: extract insights gained from clustering to support agricultural decision making, including crop selection, resource allocation, and risk management. By translating complex weather data into actionable recommendations, this study can contribute to more sustainable and productive agricultural practices.
- Integration [9]: There are many tools to help agricultural stakeholders, so another possible gain from clustering is to use it in combination with other sources of information (maps, models, databases), such as the Mapbiomas³ project. The combination of artifacts produced by more than one tool can significantly contribute to increasing the accuracy of recommendations and predictions.

The use of SIMAGRO-RS data as in this work, the first carried out on this dataset, and the publication of its results is important so that more people, especially researchers and producers, can explore the information it makes available and benefit from it. Therefore, computer science researchers can help farmers and agrometeorologists with the necessary technologies to read and interpret data, transforming it into useful information.

This research is intended to have significant implications for the advancement of Information Systems, especially in agriculture. Its findings can be applied to enhance Decision Support Systems (DSS) tailored for farmers, providing more precise and context-aware recommendations based on the clustering of meteorological data. Also, this paper contributes to the development of customized agricultural information platforms that integrate clustering algorithms, aligning with the trend toward user-centric and adaptive solutions in information systems. The case study presented here supports the evolution of the proposed methodology to a framework by incorporating meteorological clustering into many agricultural practices, such as irrigation scheduling, and pest control. Its insights can be integrated into farm management software, enhancing decision-making modules for crop planning, resource allocation, and risk management. Furthermore, the research encourages the integration of the SIMAGRO-RS information systems with other systems and technologies, such as Internet of Things (IoT) devices and sensors, ensuring real-time data feeds for continuous improvements.

³MapBiomas Brasil — <https://brasil.mapbiomas.org/>

6 CONCLUSION AND FUTURE WORKS

This work proposes a clustering of meteorological stations maintained by the SIMAGRO-RS project, in the state of Rio Grande do Sul, in southern Brazil. Variables Temperature, Relative Humidity, Rainfall, and Solar Radiation collected by 19 meteorological stations located in different municipalities in Rio Grande do Sul were analyzed. The dataset was extracted from climate data for the last four years, over the last three seasons of agricultural production in the state, between the months of October and March. After extraction, filtering and quality checking of the data were performed, and they were aggregated into daily values (for Temperature and Relative Humidity, the average was used, and for Rainfall and Solar Radiation, the accumulated value of the day was used). After aggregation, the data were also standardized to scale its representation uniformly.

The generated multivariate time series were then clustered using the K-means, K-medoids, and Hierarchical Agglomerative Clustering algorithms. To define the optimal number of clusters k , the methods *Elbow Plot* and *Silhouette Score* were used, analyzed in a complementary way. The analysis indicated the value of $k = 2$. The data were then clustered and the results presented and discussed. A clustering evaluation method based on supervised learning (linear regression) was proposed to identify the algorithm that resulted in the most appropriate clusters. The best result was then obtained using the K-medoids algorithm. A software package was developed and made available to encourage continued research in this direction and with this data source (SIMAGRO-RS). Finally, suggested uses of this clustering produced were listed.

As future work, the authors intend to deepen the analysis of climate variables, in different ways, including increasing the number of variables, investigating appropriate strategies to resolve missing, null and outliers, exploring the performance and results of other clustering algorithms, and analyzing the impact of meteorological phenomena on clustering, such as the ENSO (El-Niño Southern Oscillation), which has been directly impacting climate change in several places around the world, especially in southern Brazil. An expected impact is the use of the results of this work in agricultural applications, especially aimed at regional crops in RS. More broadly, a possible future direction of work is also the integration of the methodology proposed here and SIMAGRO-RS databases with other platforms, such as some mentioned previously.

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